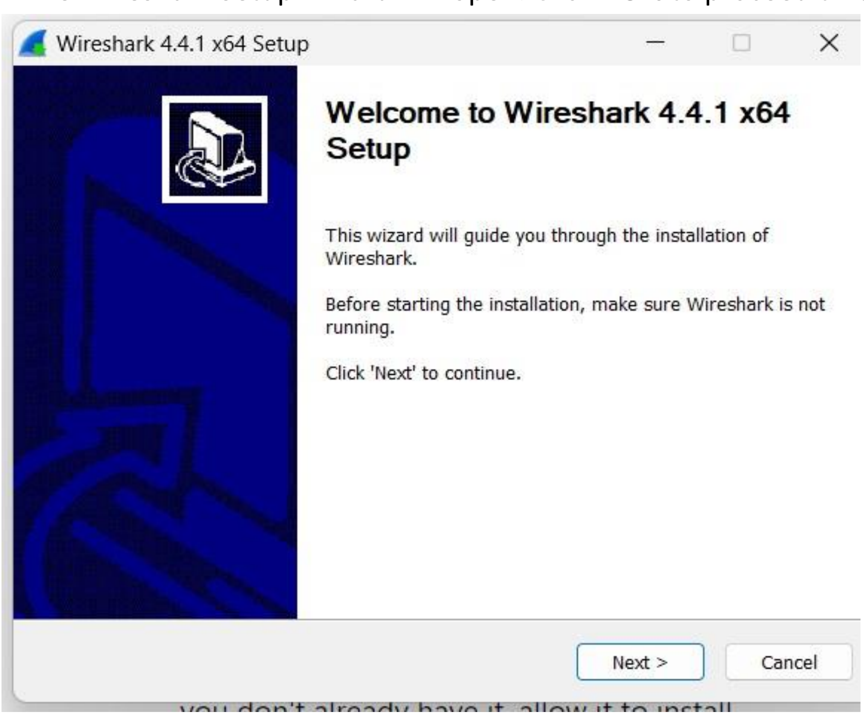
First Tool: Wireshark

Visit the Wireshark Website

Go to the official Wireshark download page:https://www.wireshark.org/download.html Choose the **Windows Installer (64-bit)** option to begin the download.

Once the download is complete, locate the downloaded .exe file and double-click it to run the installer.

The Wireshark Setup Wizard will open. Click **Next** to proceed through the installation process.

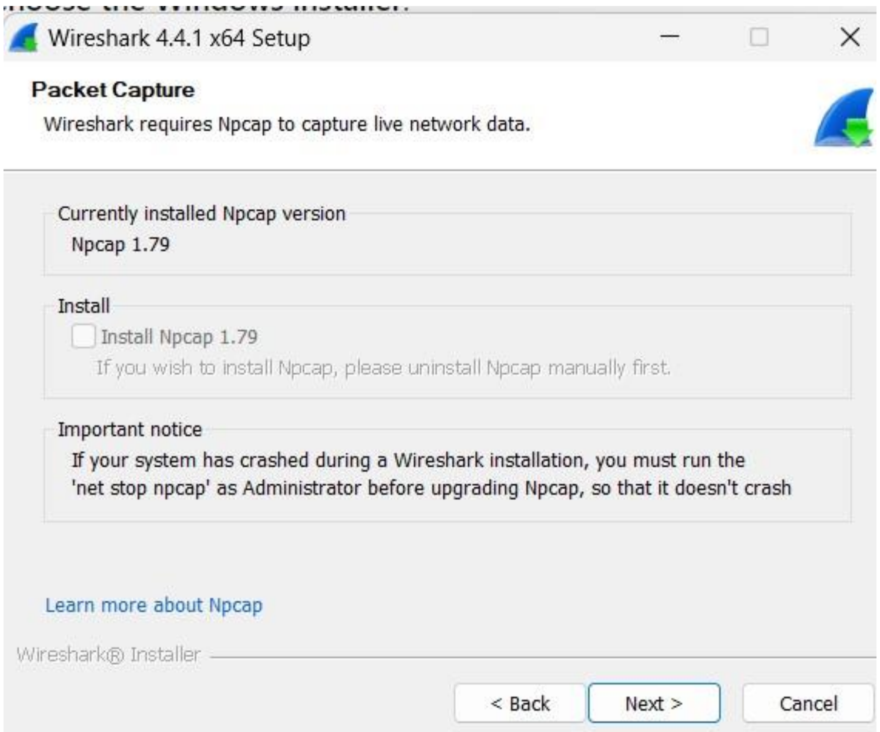


Continue clicking **Next** until you reach the option to choose the installation location. Select your preferred location and click **Next**.

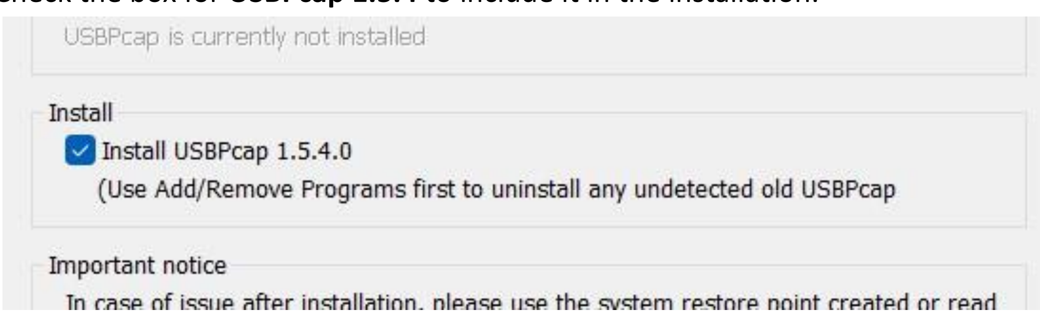
During the installation, you will be prompted to install **WinPcap**, which is used for packet capturing. If you do not already have it installed, allow it to install.

I already have Npcap 1.79, so there's no need to download it. But if you don't have it, make sure to check the option to download it.

If you have **Npcap** version 1.79 or later, you can skip this step. Otherwise, ensure the option to download and install Npcap is checked.



Check the box for **USBPcap 1.5.4** to include it in the installation.

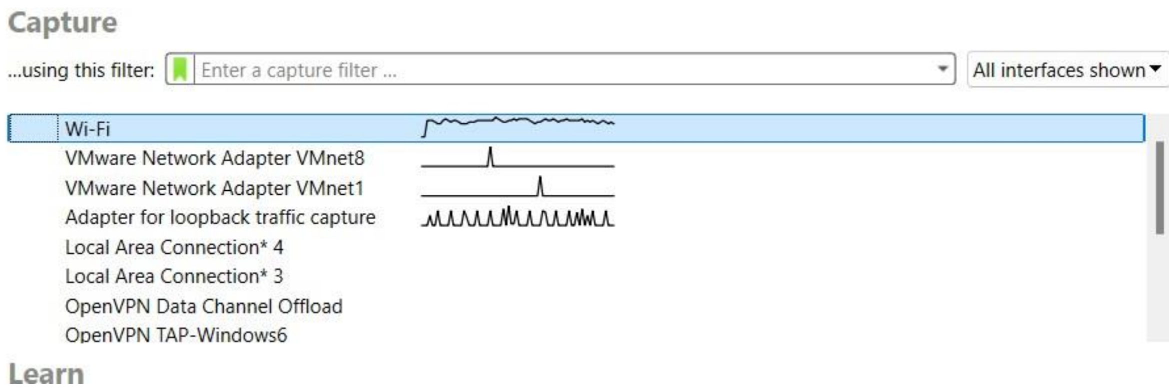


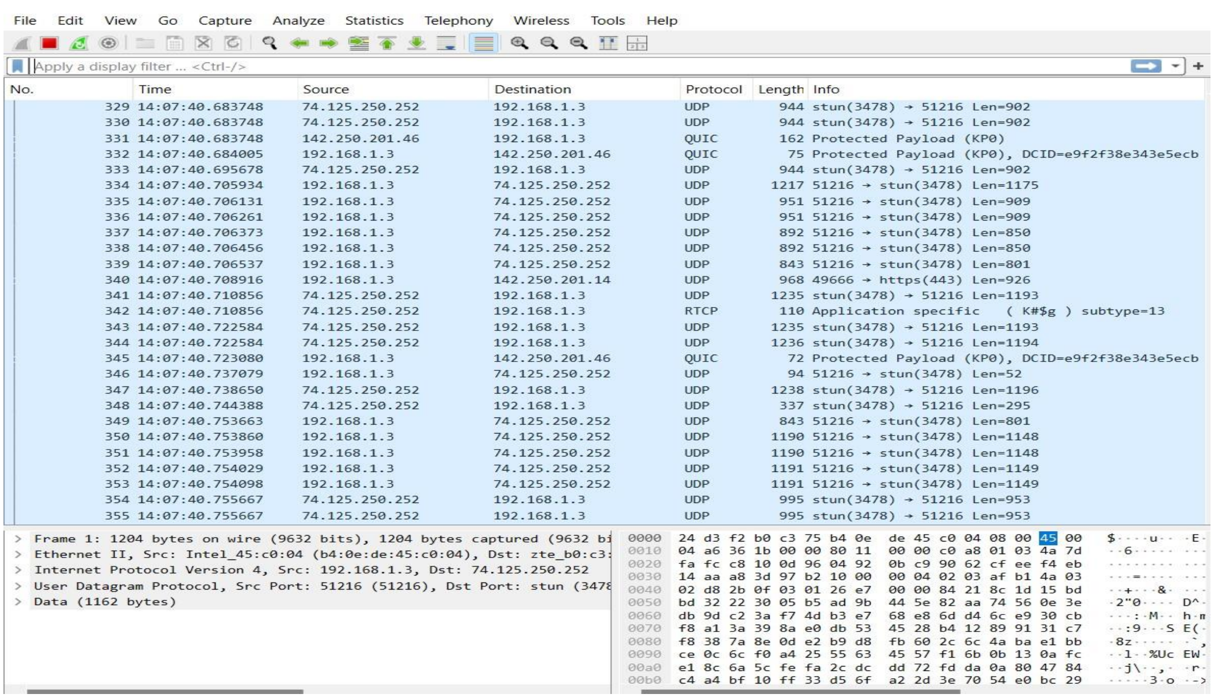
After following the above steps, click **Finish** once the installation is complete. Your system may need to restart.

Capturing Packets

Launch Wireshark from your Start Menu or desktop shortcut.

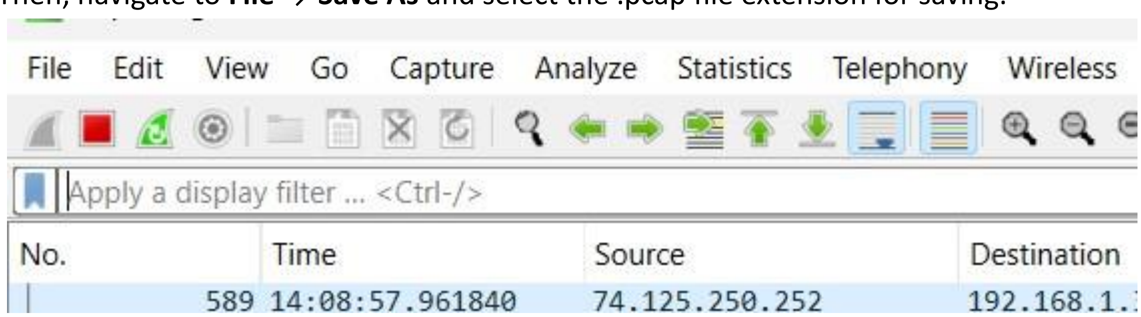
Once opened, Wireshark will begin capturing packets automatically. You can select packets transmitted over Wi-Fi.



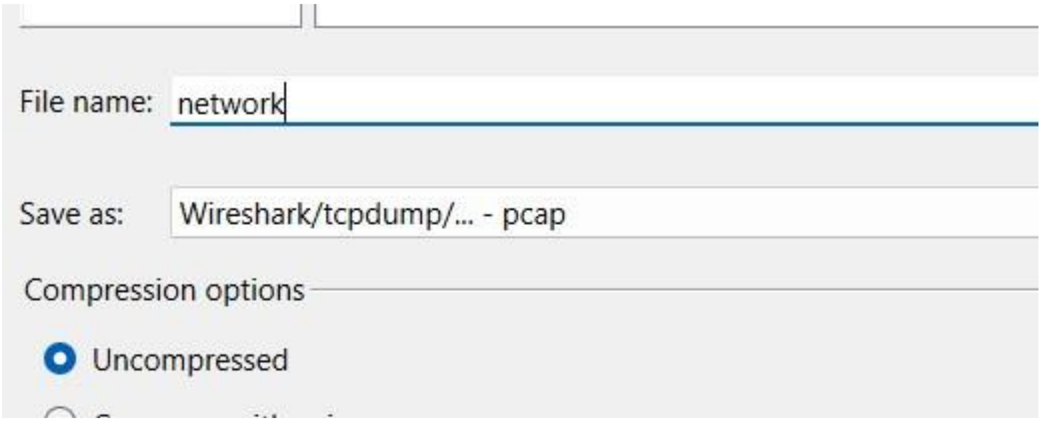


To save the captured packets in a .pcap file, you must first stop the capture by clicking the red stop button.

Then, navigate to **File** → **Save As** and select the .pcap file extension for saving.



Then file❼ save as❼ choose pcap extension

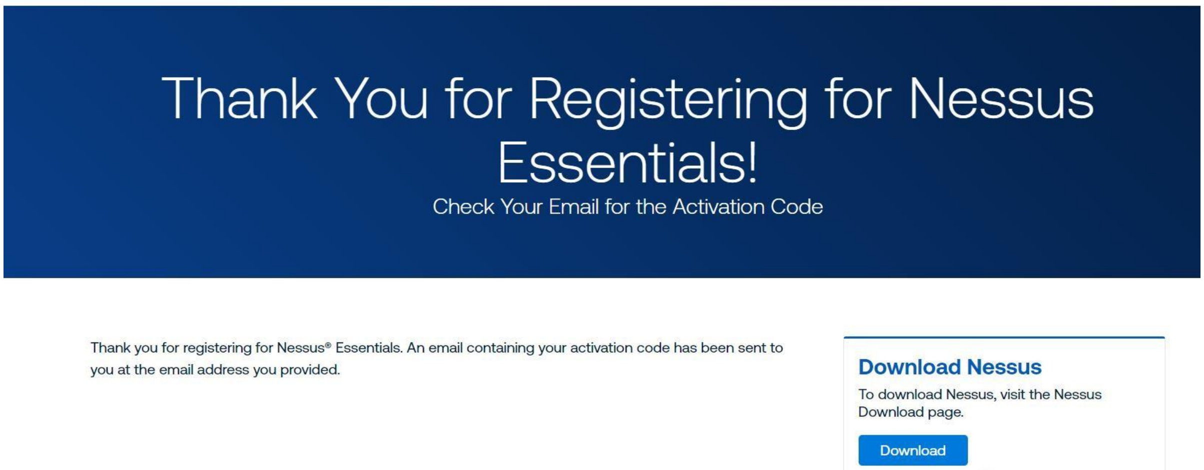

File name:
network
Save as:
Wireshark/tcpdump/... - pcap
Compression options
Uncompressed

Second Tool: Nessus Tool Installation

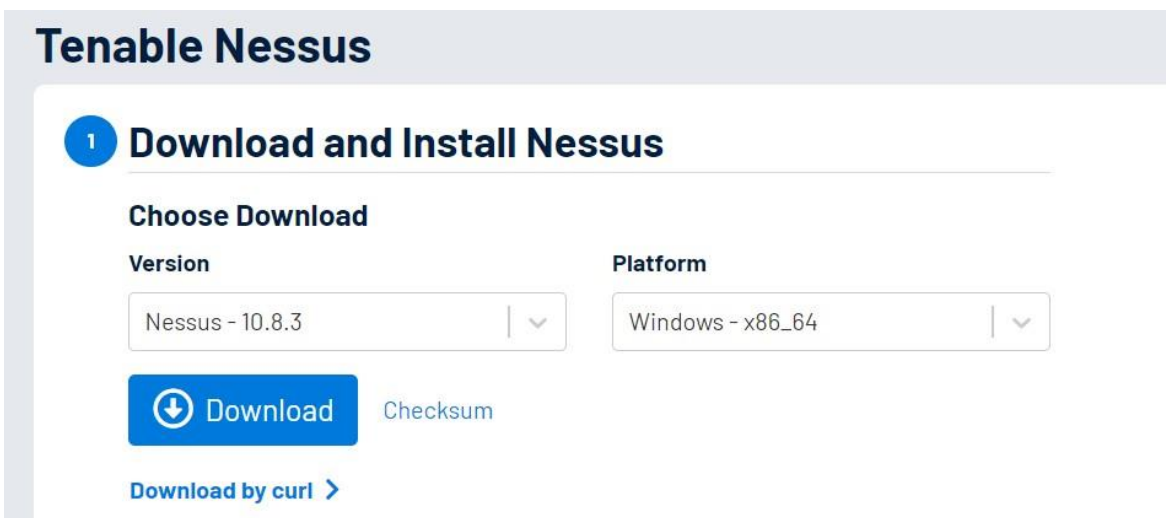
Steps

o to the official Nessus download page: https://www.tenable.com/products/nessus/nessus-essentials

Enter your information to register for Nessus. After completing the registration, check your email for an activation code. Save this code for later use.



Select the appropriate version of Nessus for your system and begin the download.



Once the download is complete, double-click the installer to launch the Tenable Nessus.

Click **Next**.

Accept the license agreement and click **Next** again.

Choose your preferred installation location and click **Next**.

Click **Install** to begin the installation process.

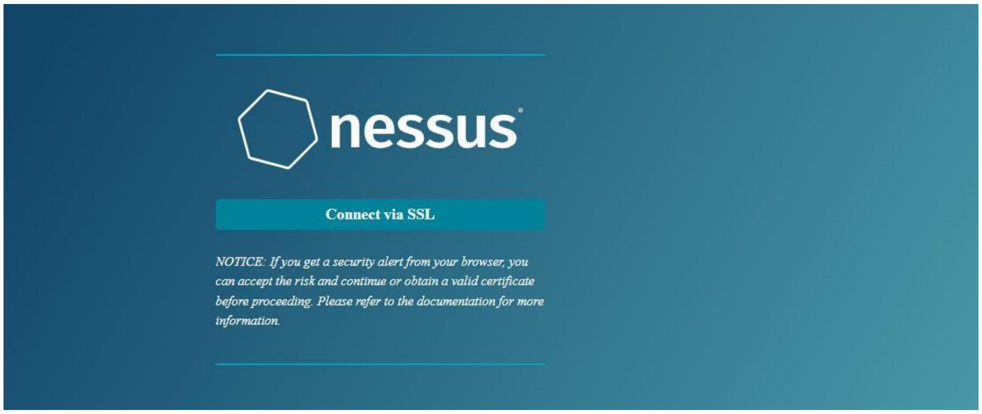
Once installed, Nessus will start running as a service.

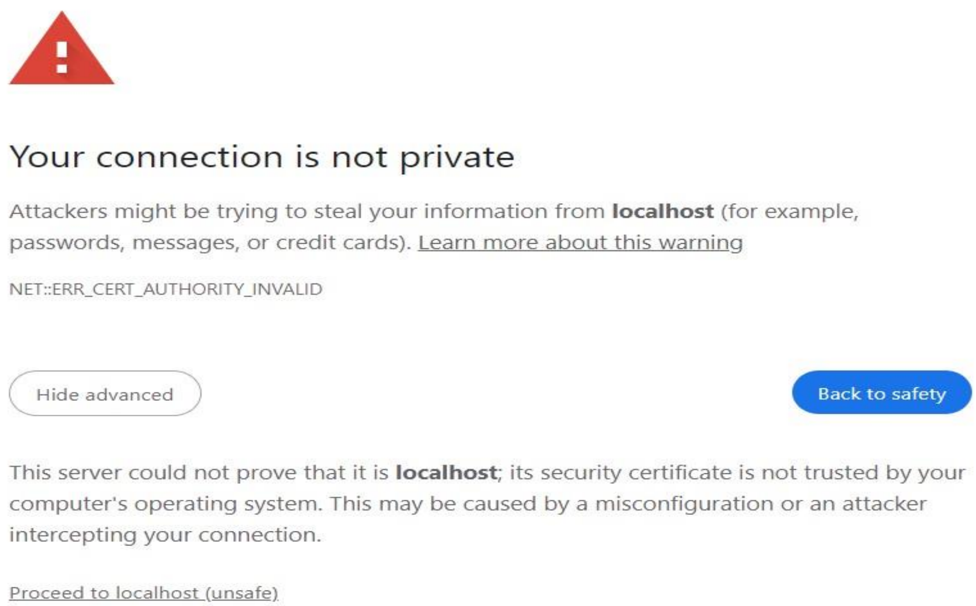
Once the installation is complete, click **Finish**. Nessus will start running as a service automatically.

Accessing the Nessus Web Interface

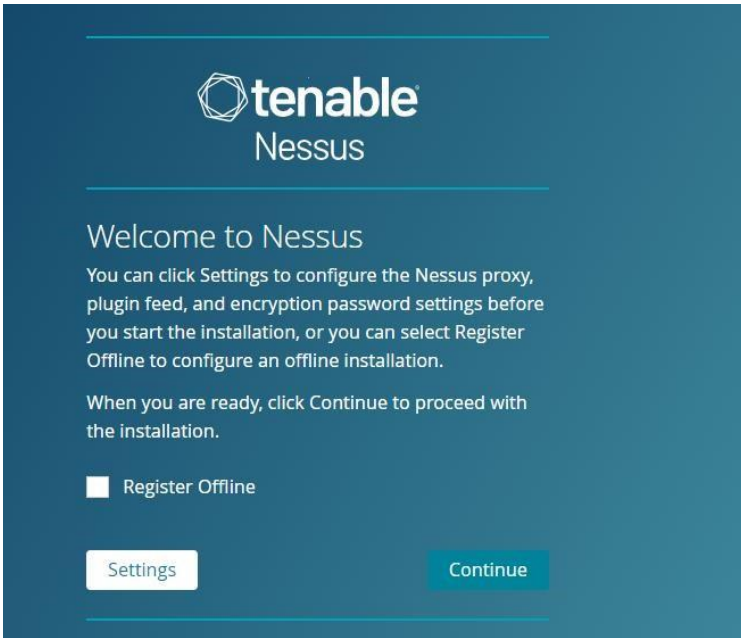
Navigate to https://localhost:8834/ to access the Nessus web interface. If the interface does not open automatically, you can enter this URL manually.

When prompted with a "Connection is not private" warning, click Advanced, then select Proceed to localhost.

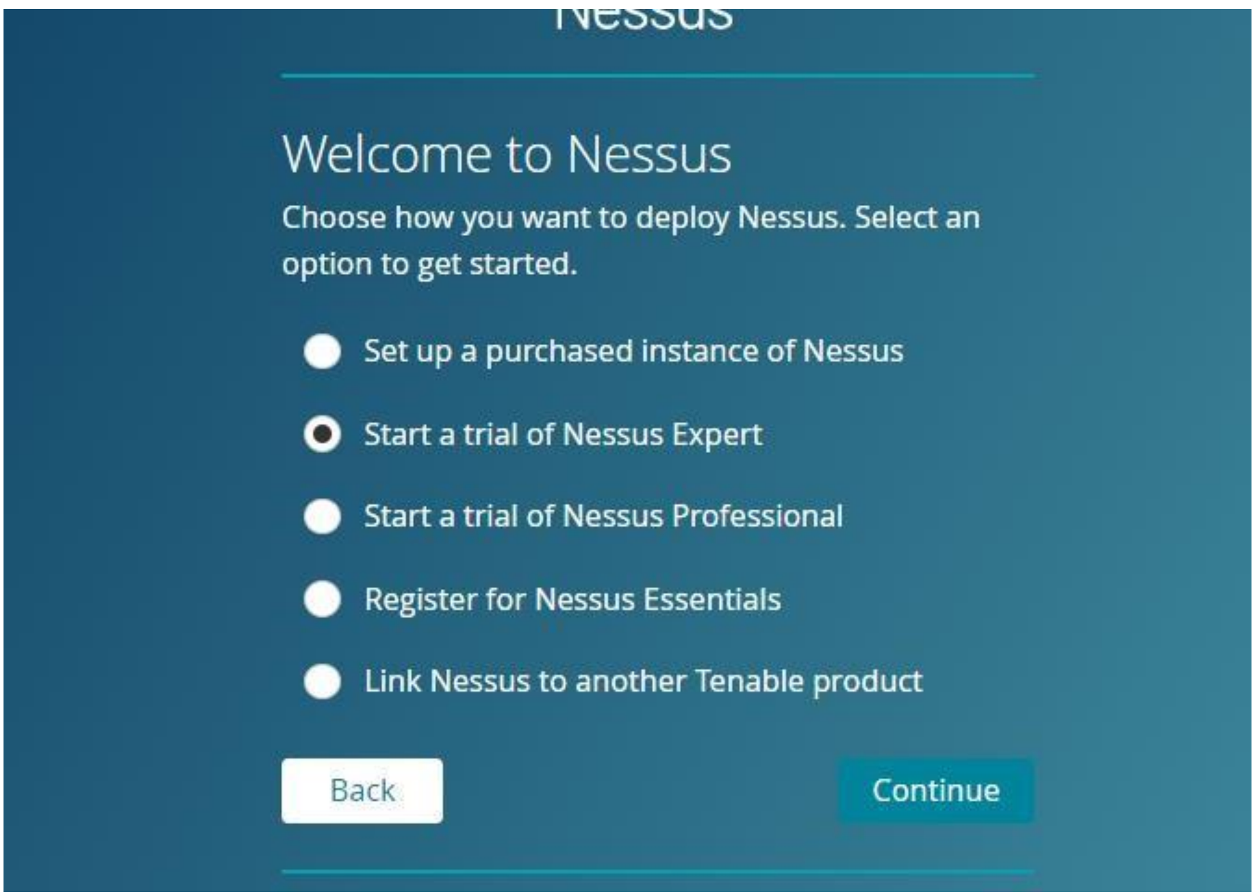




Press continues



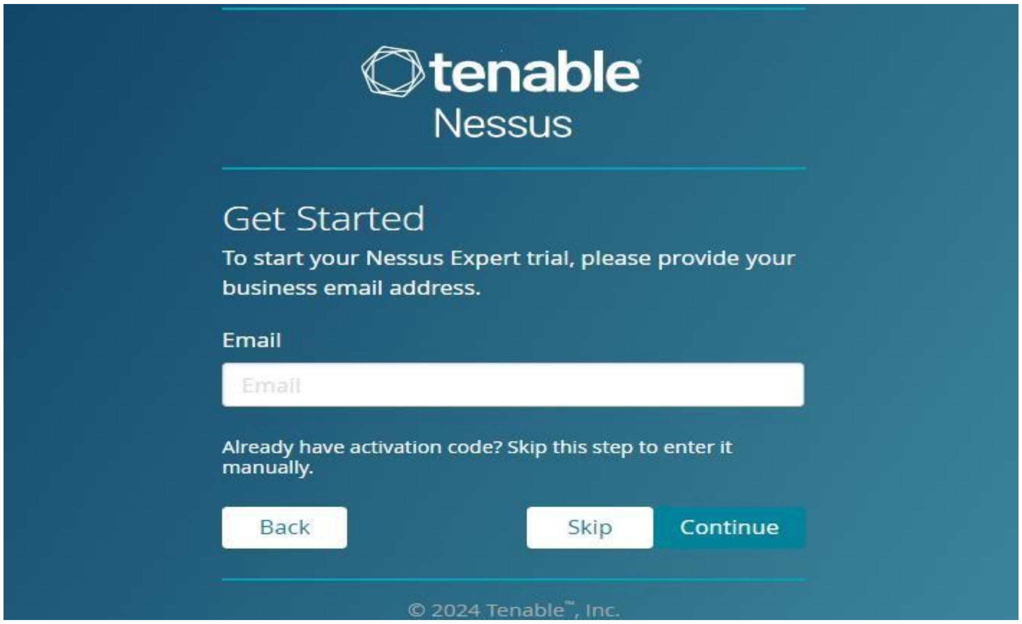
Then start trial of nessus expert, then continue

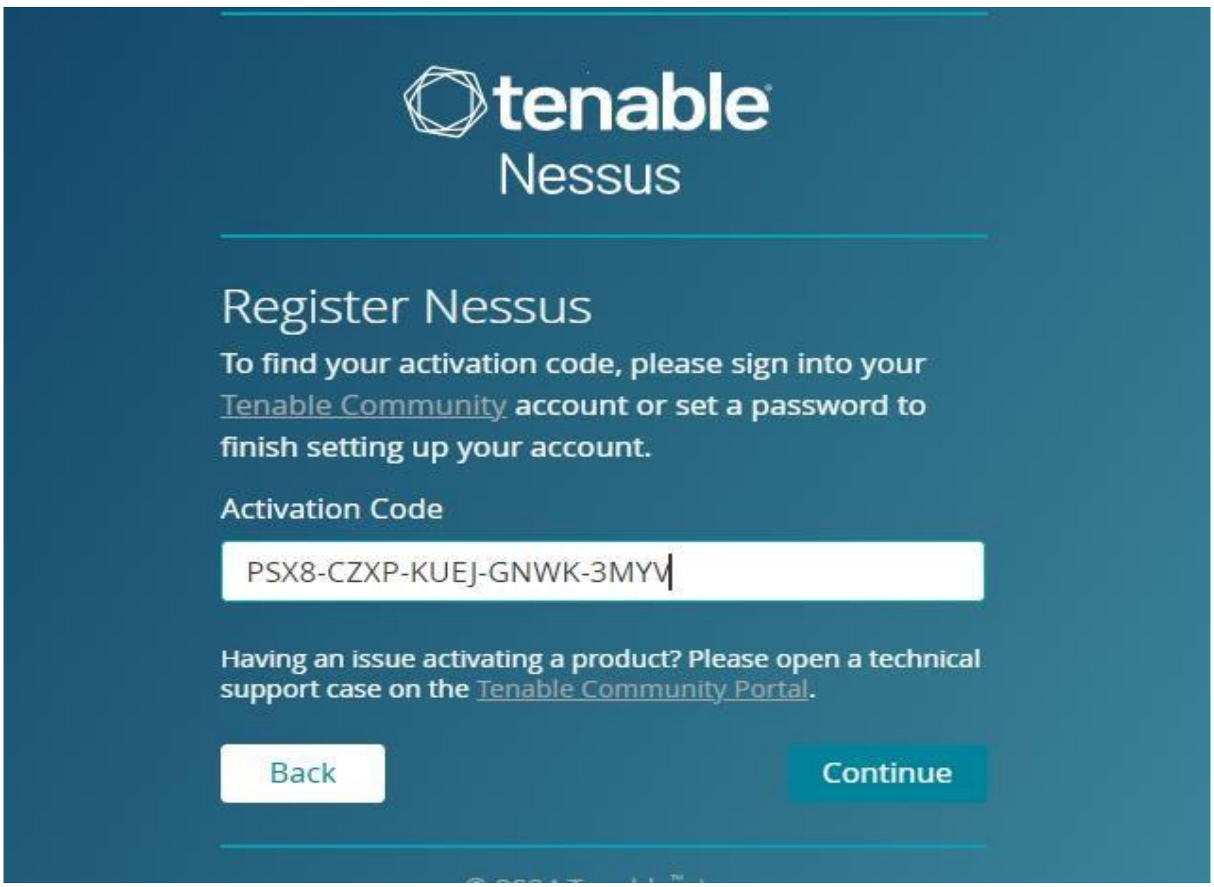


You can skip the email registration step, as you have already completed it.

Enter the activation code that was sent to your email and click **Continue**.

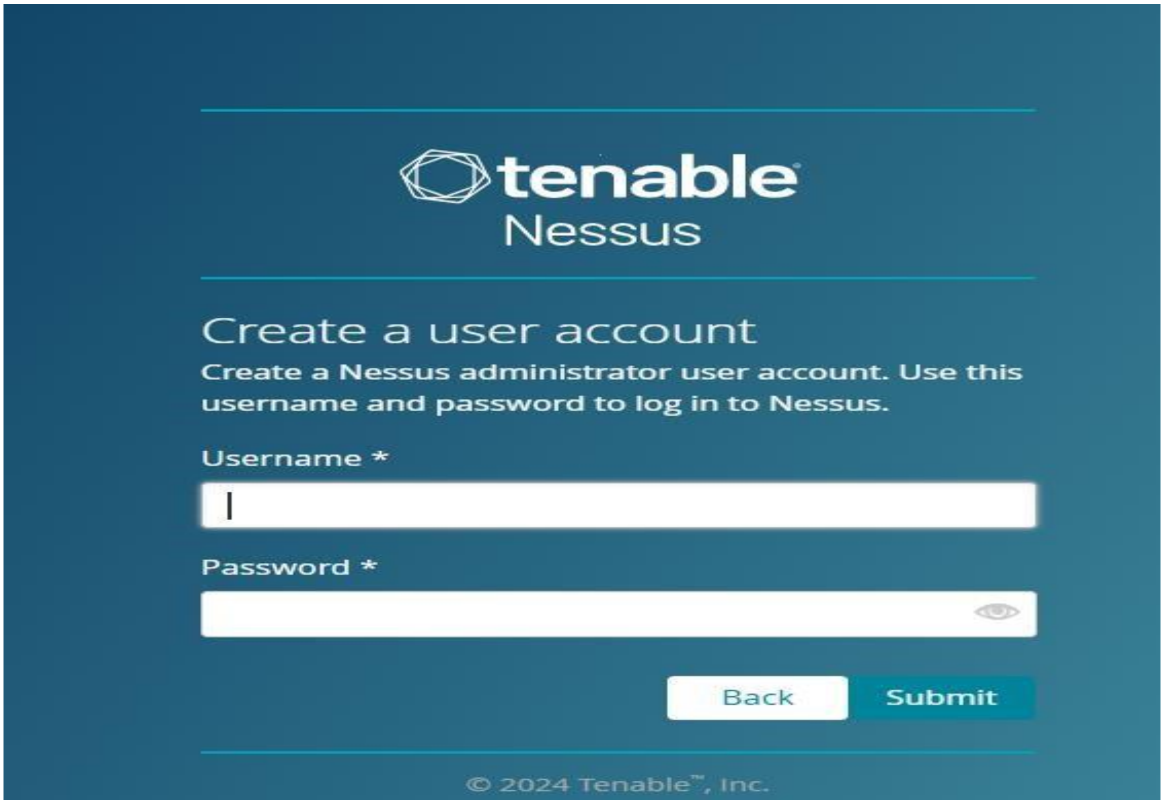
Click **Continue** again when prompted.



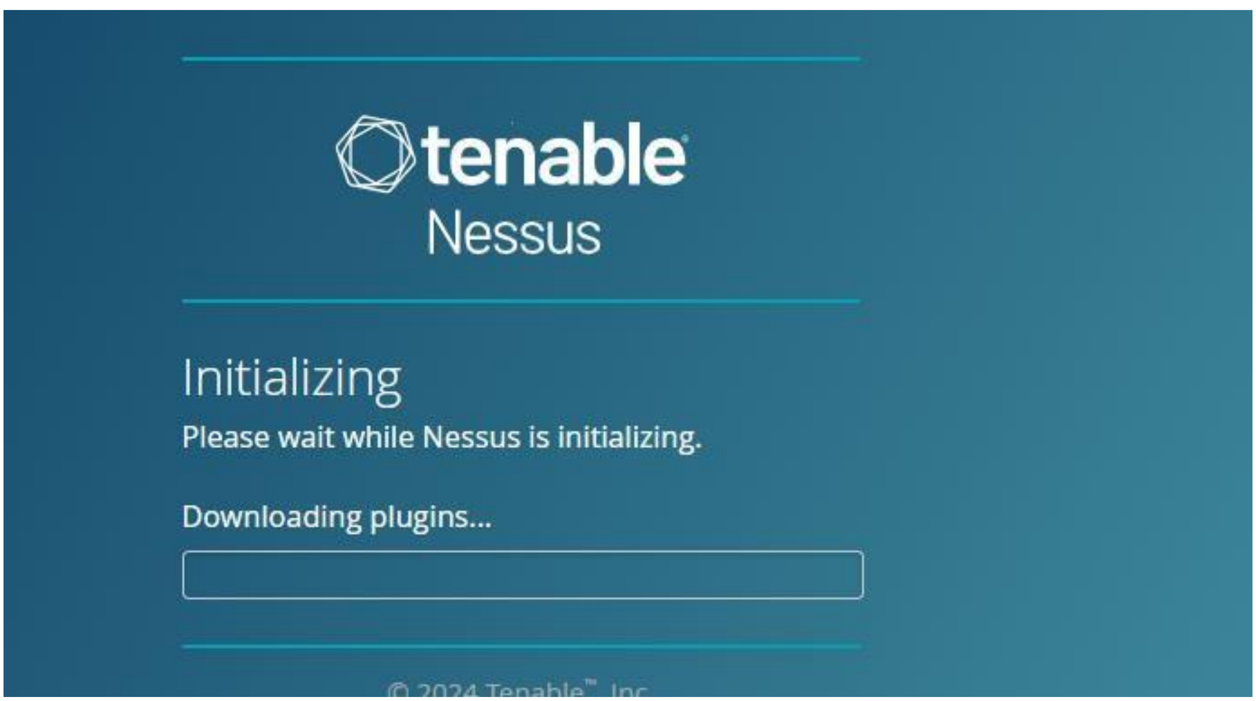


Press continue and continue again

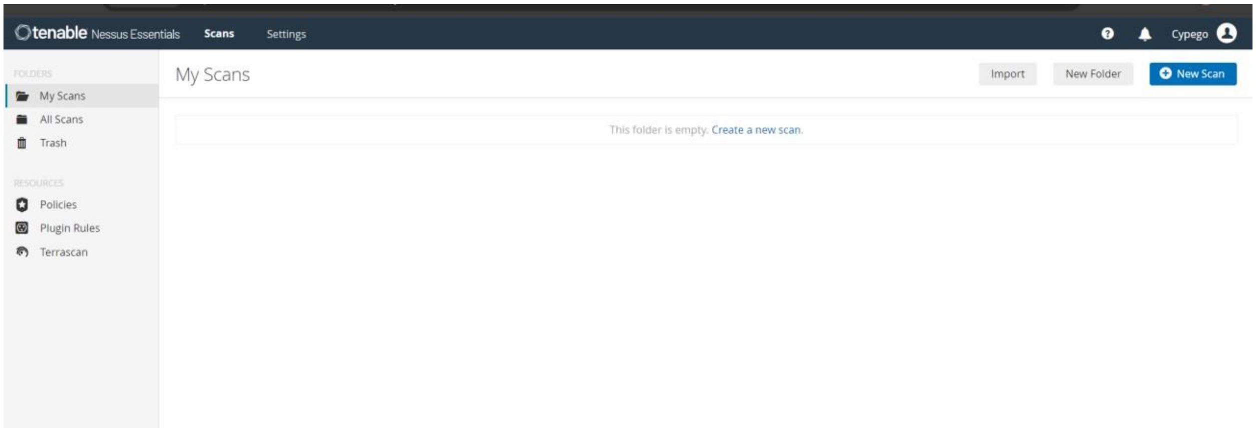
Create a username and password for accessing Nessus



Then let it to download plugins



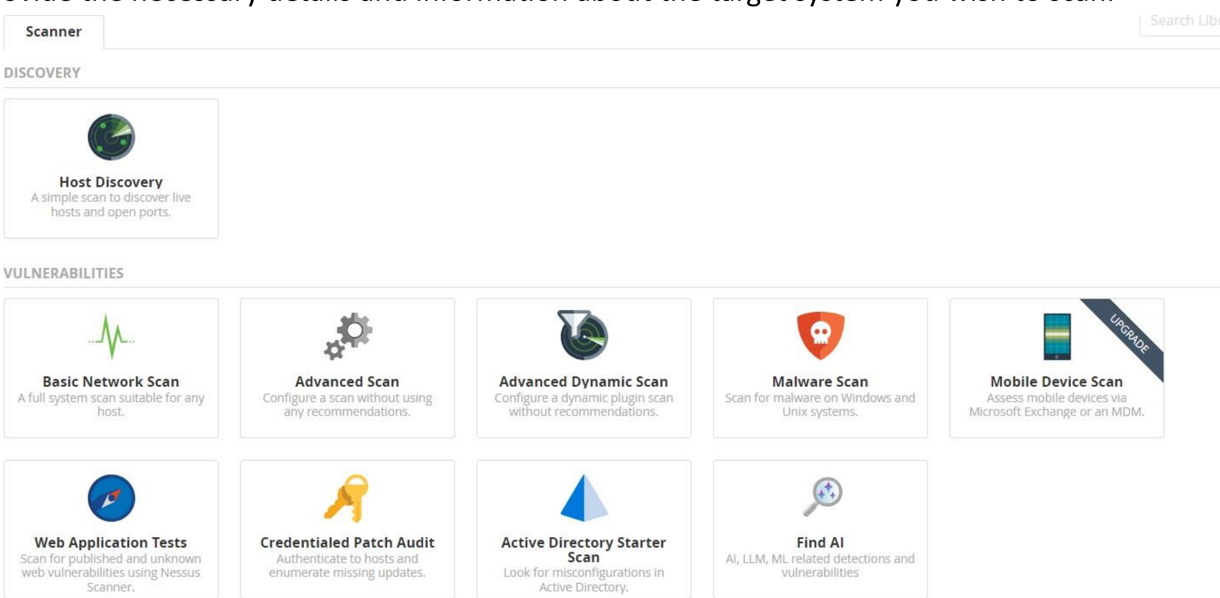
After the plugins are installed, Nessus will be ready to use.



Running Scans

From the Nessus dashboard, you can create a new scan. Choose any scan type that suits your needs.

Provide the necessary details and information about the target system you wish to scan.



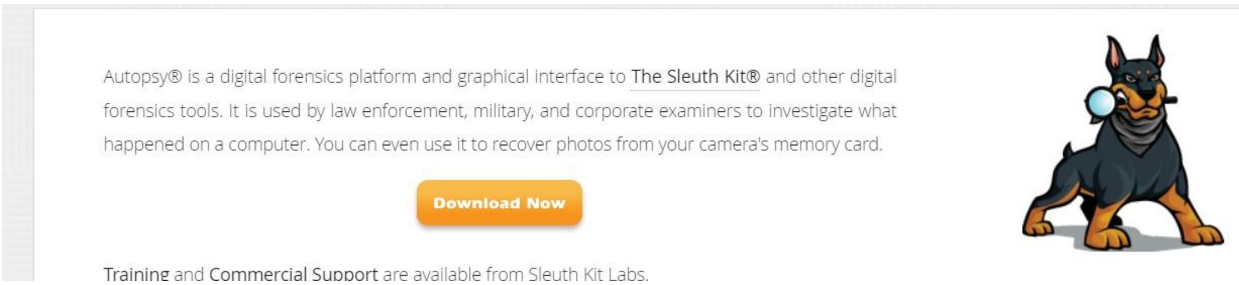
You can run scans anytime by navigating to https://localhost:8834/

Third Tool: Autopsy Tool

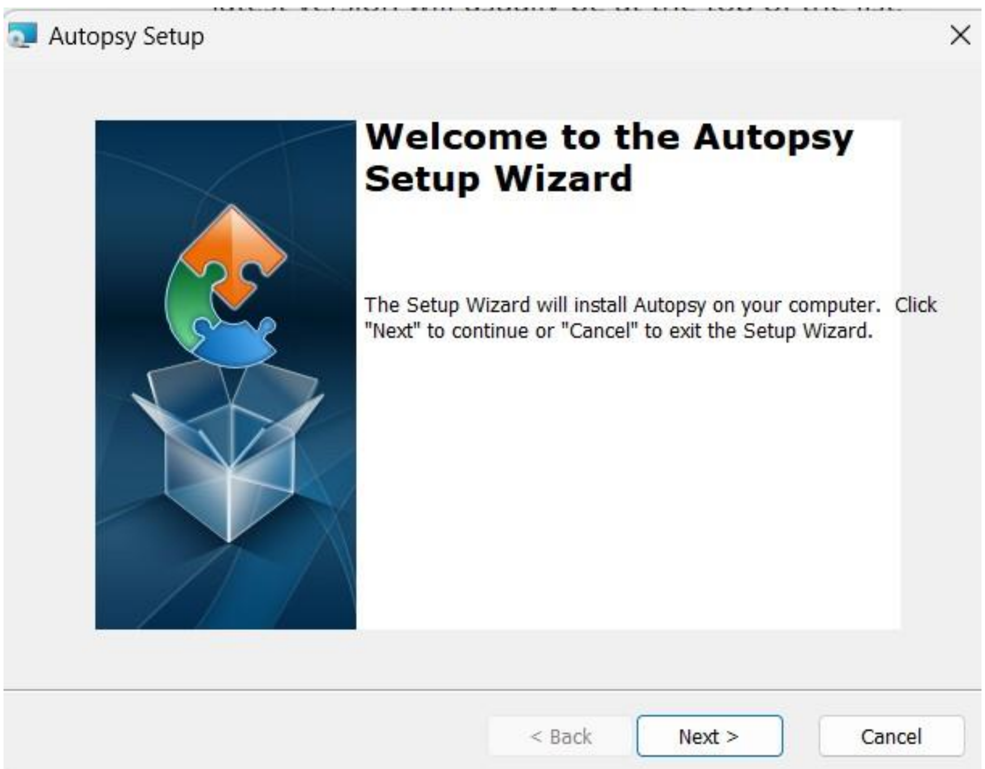
Visit the Autopsy Website

Go to the official Autopsy download page: https://www.sleuthkit.org/autopsy/ Click on

Download Now and select the appropriate version for Windows.



After the download is complete, locate the installer and double-click it to set up Autopsy.

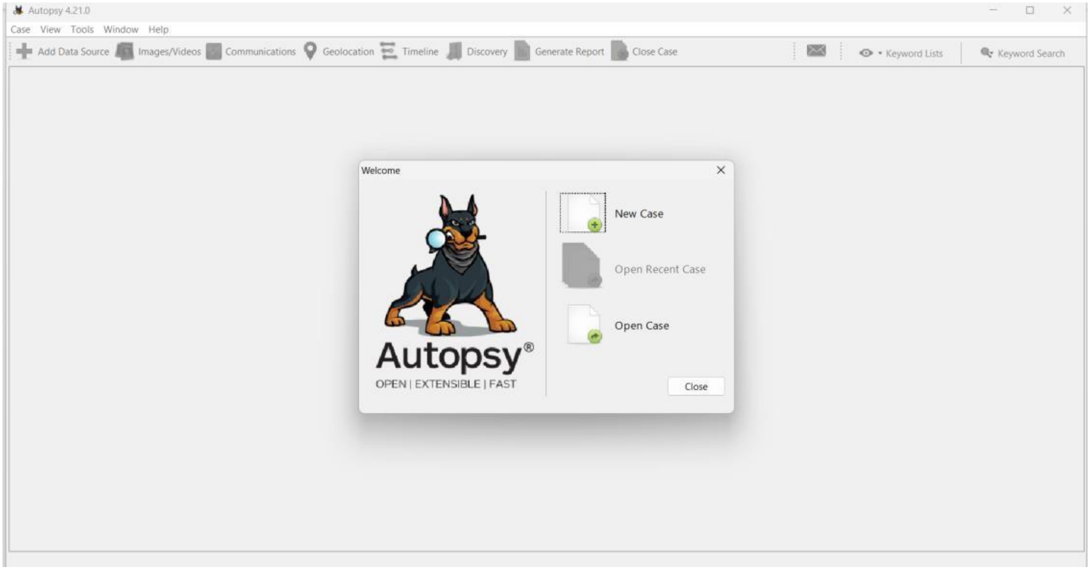


Click **Next** to proceed through the setup wizard.

Choose your preferred installation location and click **Next**.

Click **Install** to begin the installation process.

Once the installation is finished, you will have the option to create a new case or open an existing one.



Prerequisites • **Java Requirement**

Ensure that you have Java installed on your system, as Autopsy requires it to run. If you do not have Java installed, you can download it from the following link:

https://www.oracle.com/java/technologies/downloads/#java11?er=221886

Fourth Tool: Suricata

Installation Steps

Go to the official Suricata download page: https://suricata.io/download/

Look for the Windows installation package and ensure you select the appropriate version for your system architecture (32-bit or 64-bit).

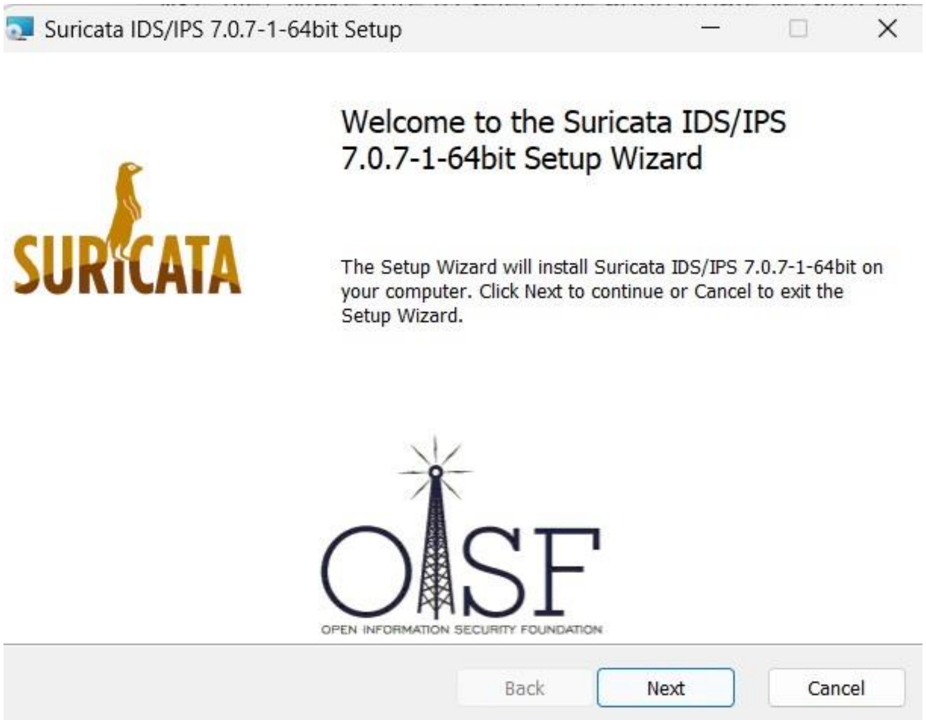
Suricata requires Npcap for packet capturing. If you do not have it installed, download it from the following link:

https://npcap.com/#download

Run the Installer: Double-click the downloaded installer file to launch the installation wizard.

Follow the Installation Instructions:

- Accept the license agreement.
- Choose the installation directory (the default location is usually fine).
- Select any additional components you want to install, if prompted.
- Click "Install" to begin the installation.
- Then finish the installation



Configuration Steps

Navigate to C:\Program Files\Suricata\.

open the suricata.yaml file in a text editor or Visual Studio.

Run ipconfig in the command prompt to find your IP address and subnet mask.

Locate the HOME_NET variable in the suricata.yaml file and set it to match your internal network.
Example: If your network is 192.168.1.0/24 (where your device has the IP 192.168.1.3), then set HOME_NET to 192.168.1.0/24.

```
vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.1.0/24, 192.168.116.0/24, 192.168.136.0/24]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
```

Ensure that any rules you want to use are not hashed (commented out) in the configuration file.

Also, ensure that the EXTERNAL_NET variable is not hashed.

```
default-rule-path: C:\\Program Files\\Suricata\\rules\\

rule-files:
 - botcc.rules
 - botcc.portgrouped.rules
 - ciarmy.rules
 - compromised.rules
 - drop.rules
 - dshield.rules
 - emerging-activex.rules
 - emerging-adware_pup.rules
 - emerging-attack_response.rules
 - emerging-chat.rules
 - emerging-coinminer.rules
 - emerging-current_events.rules
 - emerging-dns.rules
```

Also make sure externalnet is not hashed

```
EXTERNAL_NET: "!$HOME_NET"
#EXTERNAL_NET: "any"
```

Verify and Fix Rules

- Suricata uses rules to detect suspicious activity on the network. The rules are located in C:\Program Files\Suricata\rules.
- You can download additional rules from : https://rules.emergingthreats.net/open/suricata/

emerging-all.rules.zip.md5	2024-10-16T20:11:20Z	33 B
emerging.rules.tar.gz	2024-10-16T20:11:20Z	4.33 MB

Ensure that any new rules are properly declared in the suricata.yaml file.

Start Suricata

Open a command prompt and navigate to the Suricata installation directory.

To start Suricata with your IP address, enter the appropriate command in the command prompt. The logs will be stored in the log folder within the Suricata directory.

```
suricata.exe -c suricata.yaml -i 192.168.1.3
```

hint:

To test if your Suricata configuration is correct, run:

```
suricata.exe -T -c suricata.yaml
```

Fifth Tool: Volatility3

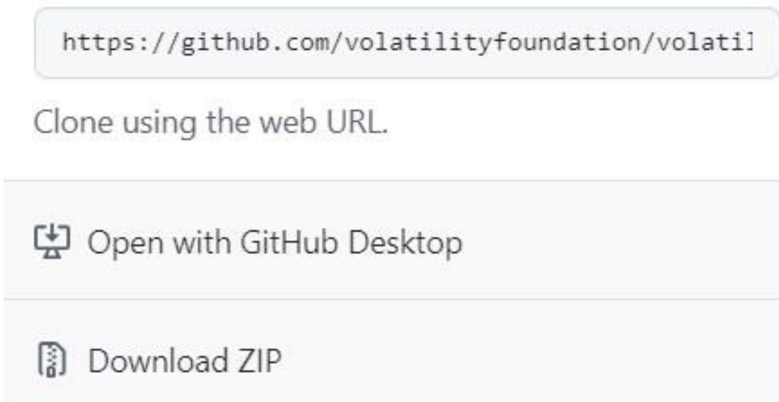
Navigate to the Volatility Repository

Visit the official Volatility 3 repository on GitHub

https://github.com/volatilityfoundation/volatility3

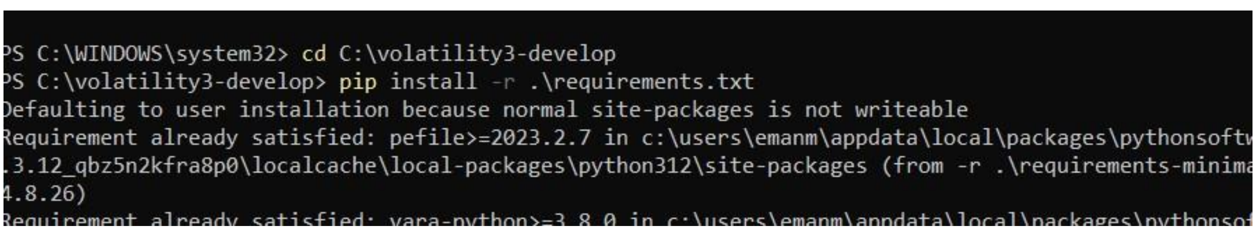
Click on the **Code** button, then select **Download ZIP**.

Once the ZIP file is downloaded, extract it to your preferred directory.



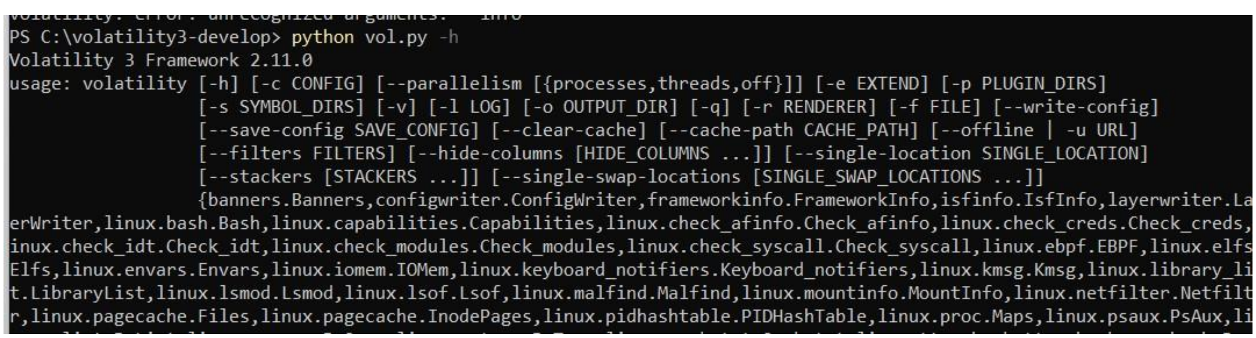
Open a command prompt and navigate to the directory where you extracted Volatility 3. Run the following command to install the required dependencies

```
pip install -r requirements.txt
```



To verify the installation and view available commands, run:

```
python vol.py -h
```



Using Volatility 3 with Memory Dumps

To use Volatility with a memory dump, execute the following command

```
python vol.py -f <path_to_memory_dump> <plugin>
```

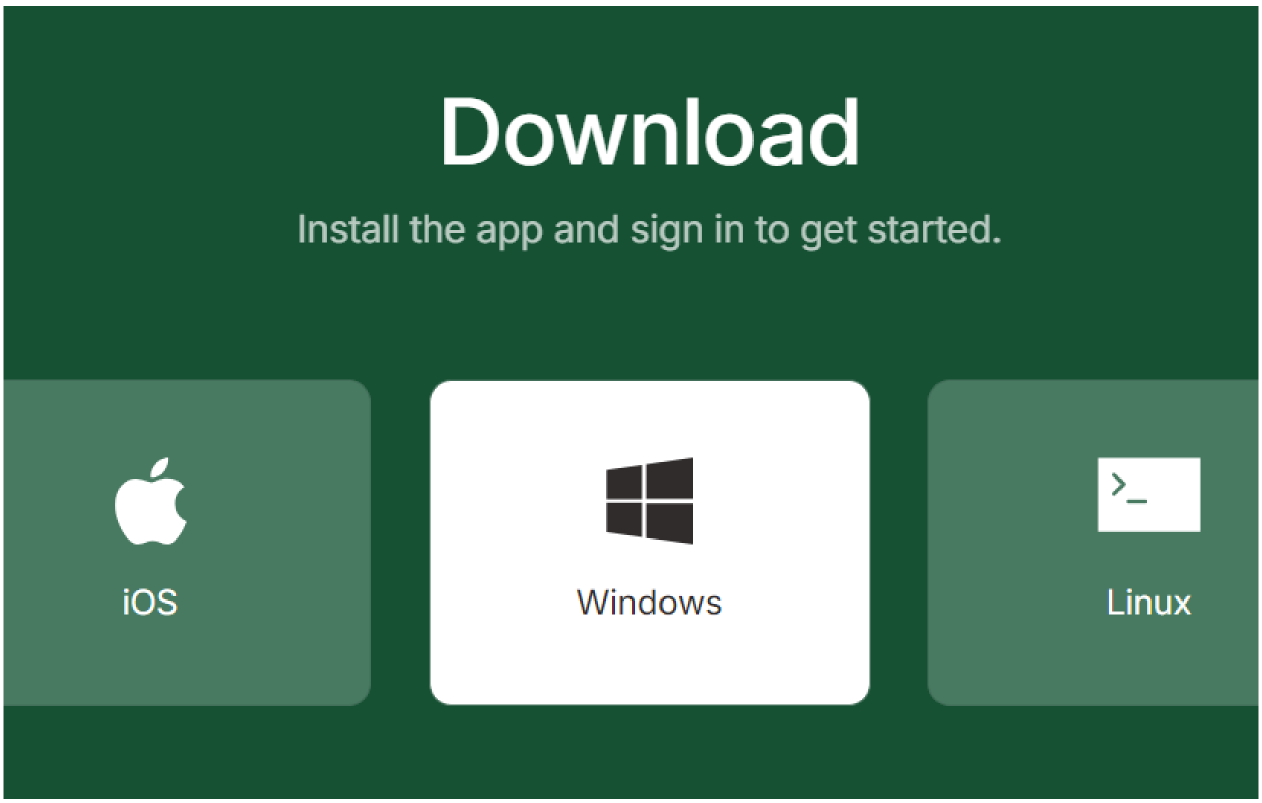
Replace <path_to_memory_dump> with the path to your memory dump file and <plugin> with the desired analysis plugin.

Sixth Tool: TailScale VPN

Download Tailscale

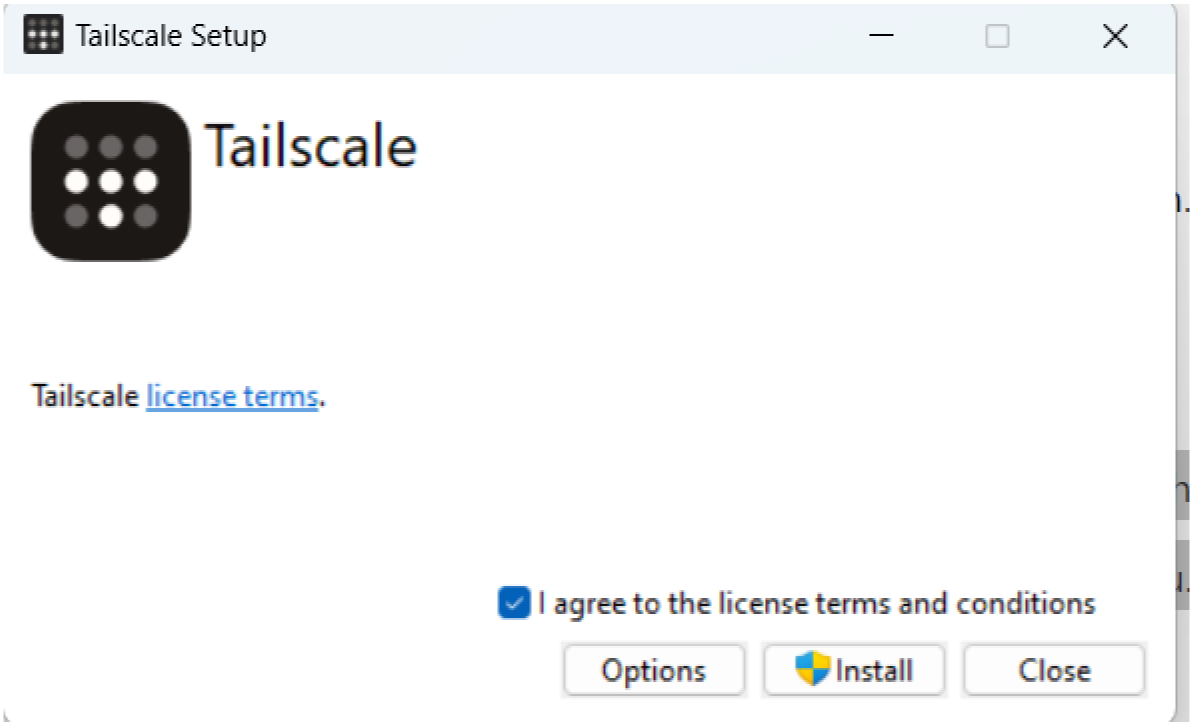
- Visit the link and click the "Download" button. Select the **Windows** option to download the installer.

  https://tailscale.com/download



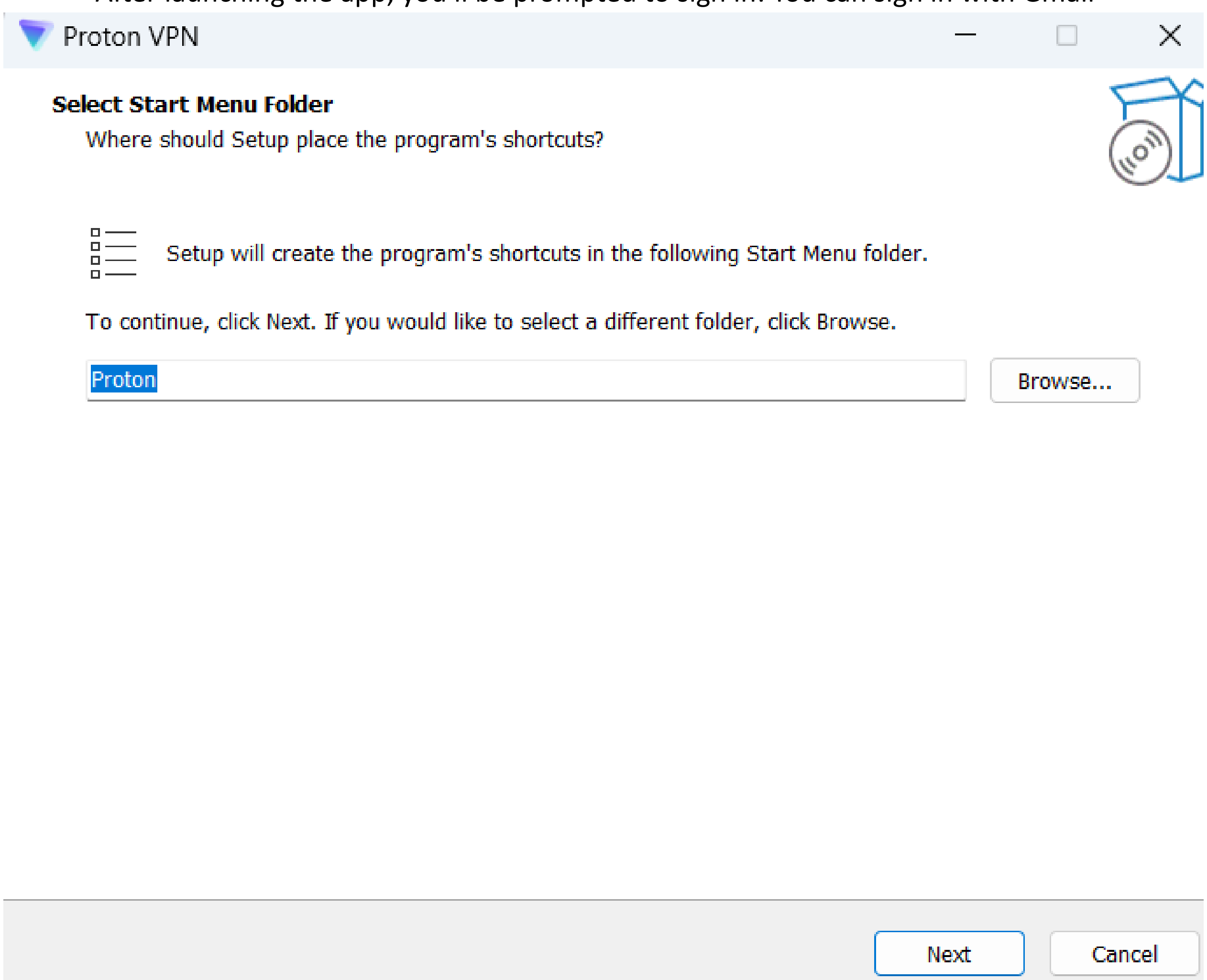
Install Tailscale

- Run the installer and follow the instructions to complete the installation process.
- Once installed, open the Tailscale app from the Start Menu.



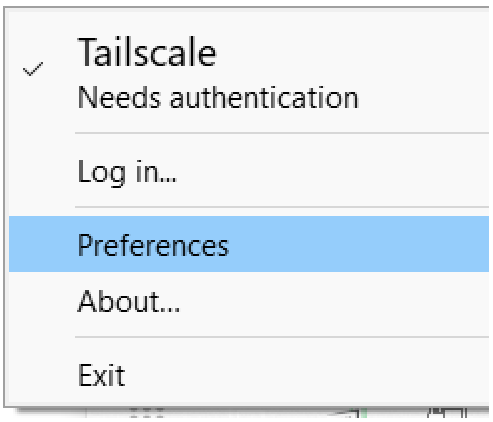
Sign In

- After launching the app, you'll be prompted to sign in. You can sign in with Gmail

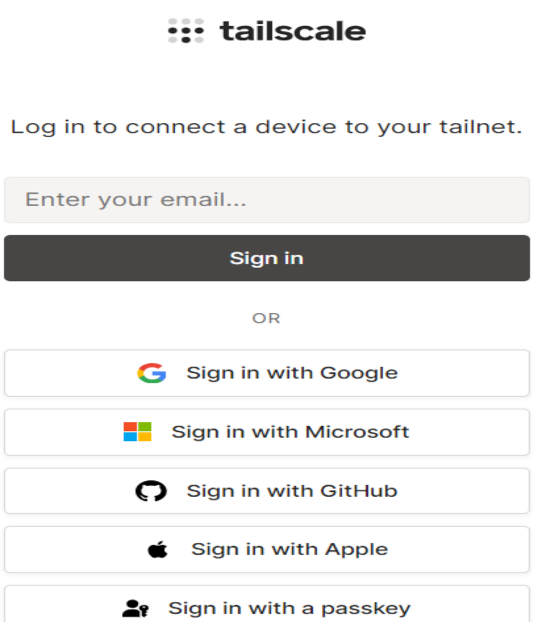


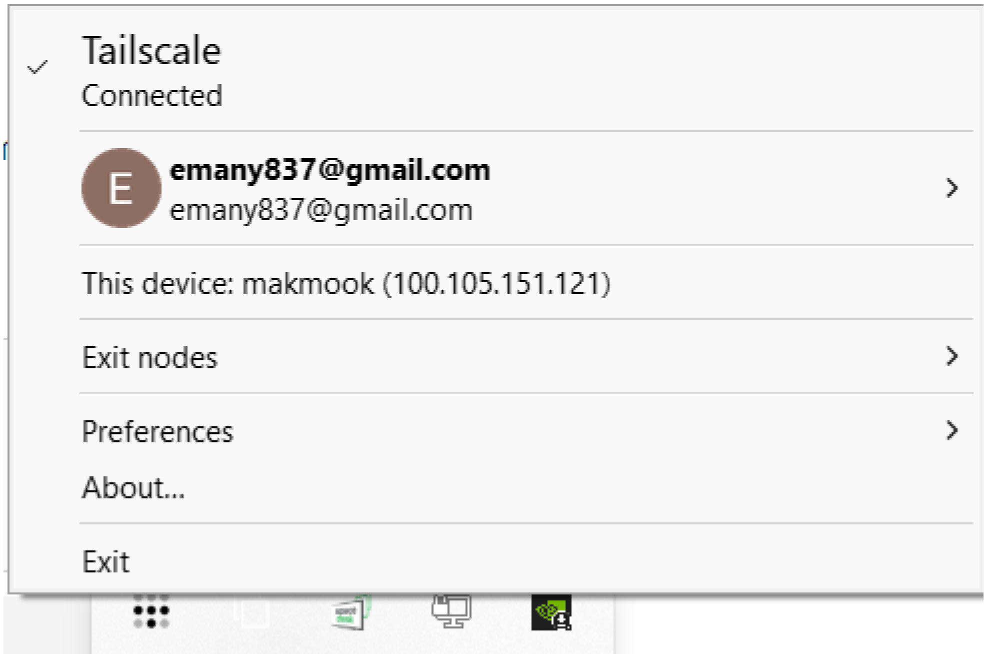
Sign In

- After launching the app, you'll be prompted to sign in. Tailscale supports multiple sign-in methods
- Try to open it from start menu, if do not appear you will find it in the bottom right corner in system tray
- Select press login



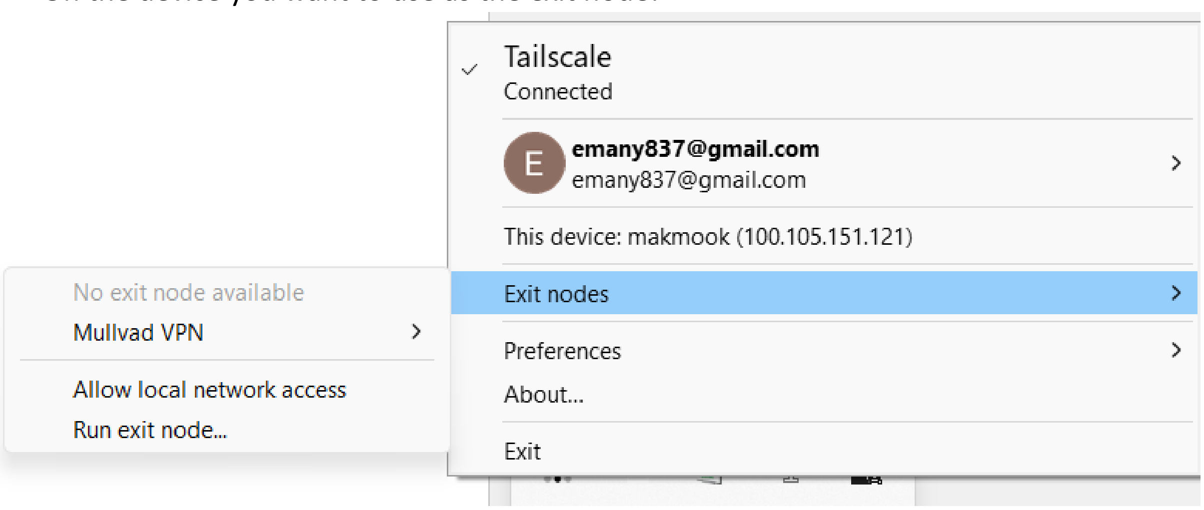
Choose method





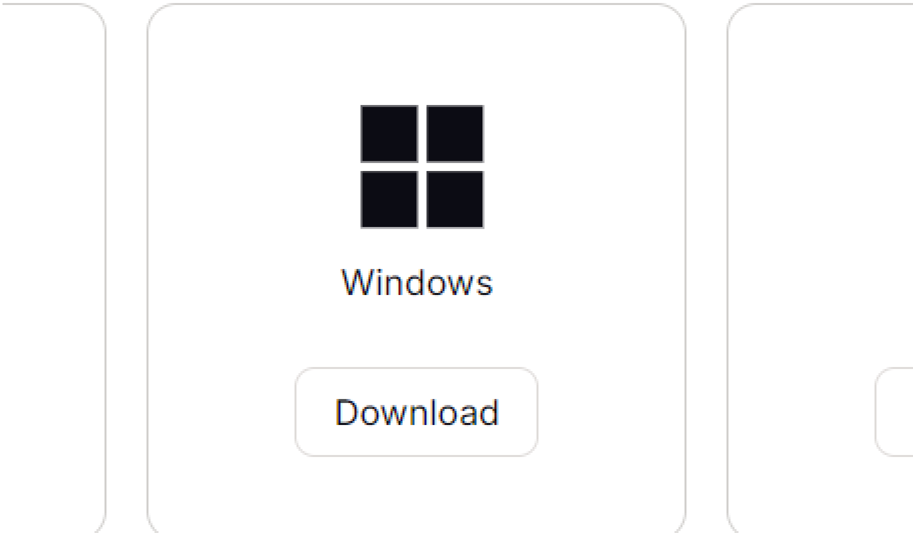
Set Up the Exit Node

- On the device you want to use as the exit node:

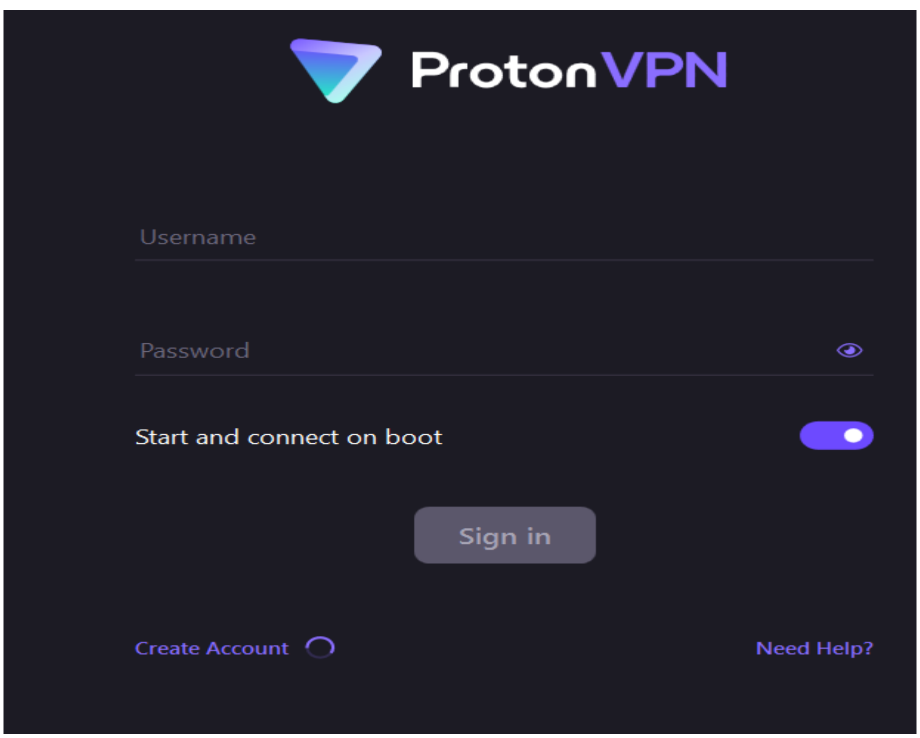


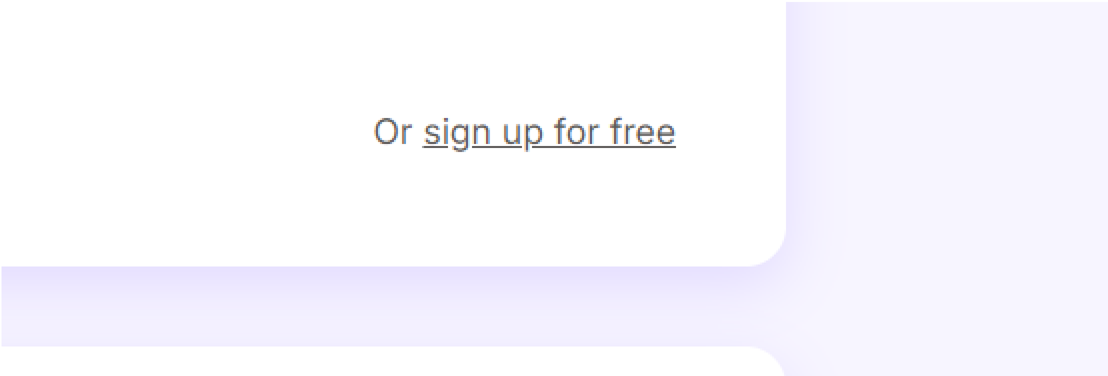
Another VPN(ProtonVPN)

Visit the ProtonVPN Website and download it

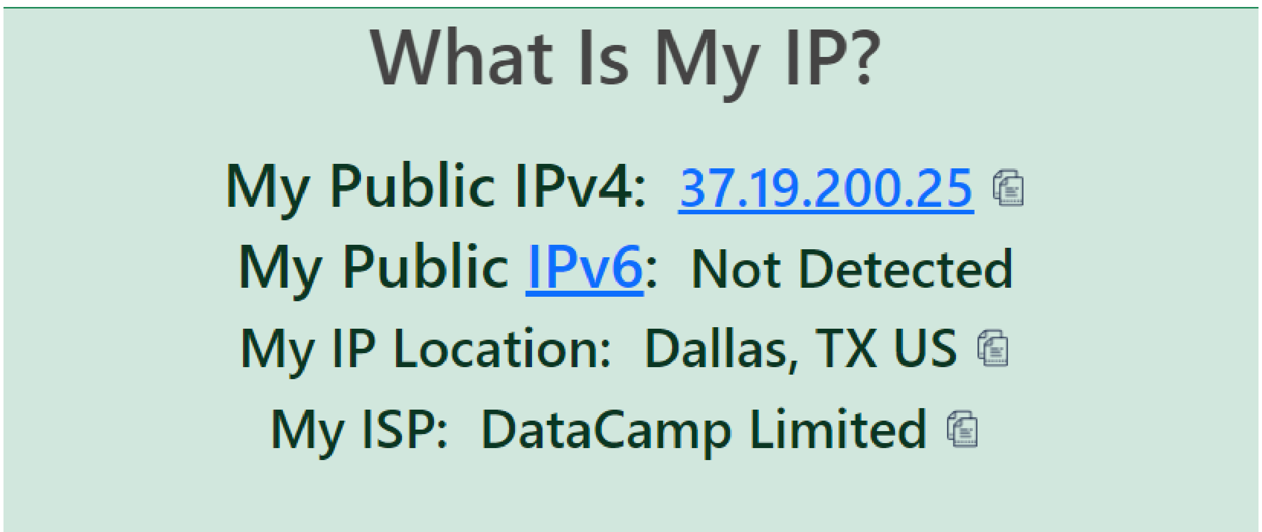


- Once the download is complete, run the installation.
- Follow the on-screen instructions to install ProtonVPN. You may need to accept the license agreement and choose the installation location.
- After installation, launch the ProtonVPN app. You will be prompted to sign in.
- If you don’t have an account, you can create one directly from the app or on the ProtonVPN website.



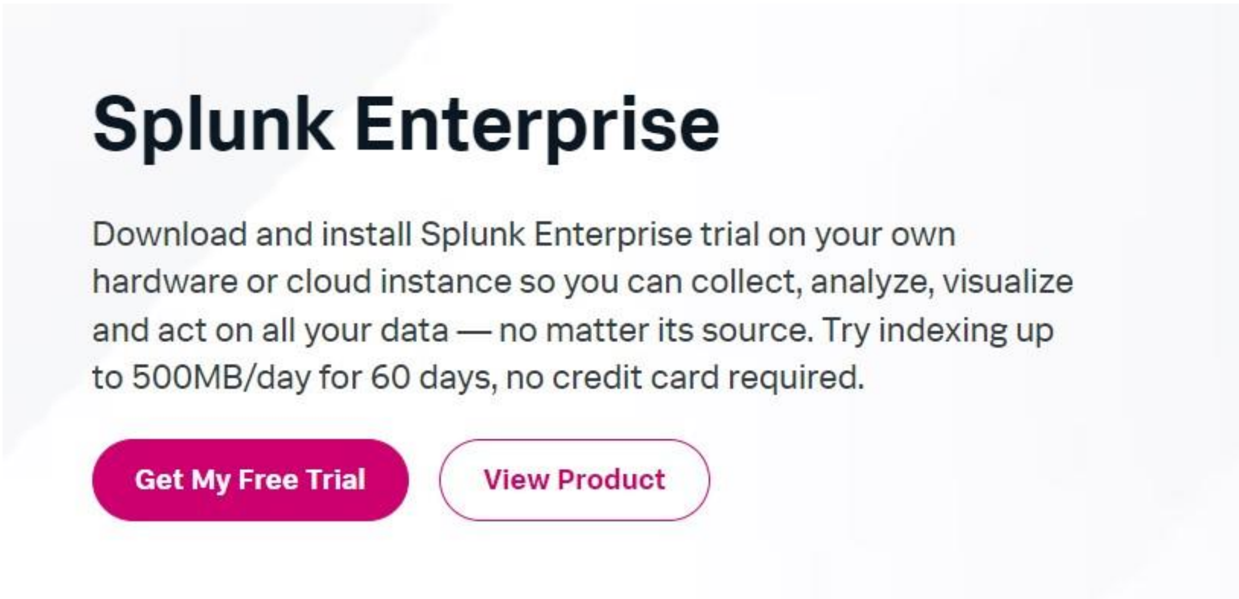


- Once signed in, you'll see a list of available servers. Choose a server location and click Connect to start using ProtonVPN.



Seventh Tool: Splunk on windows

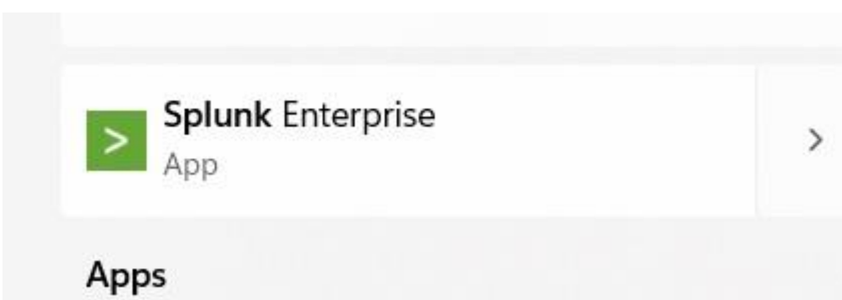
- Go to Splunk download page https://www.splunk.com/en_us/download.html
- Select splunk enterprise



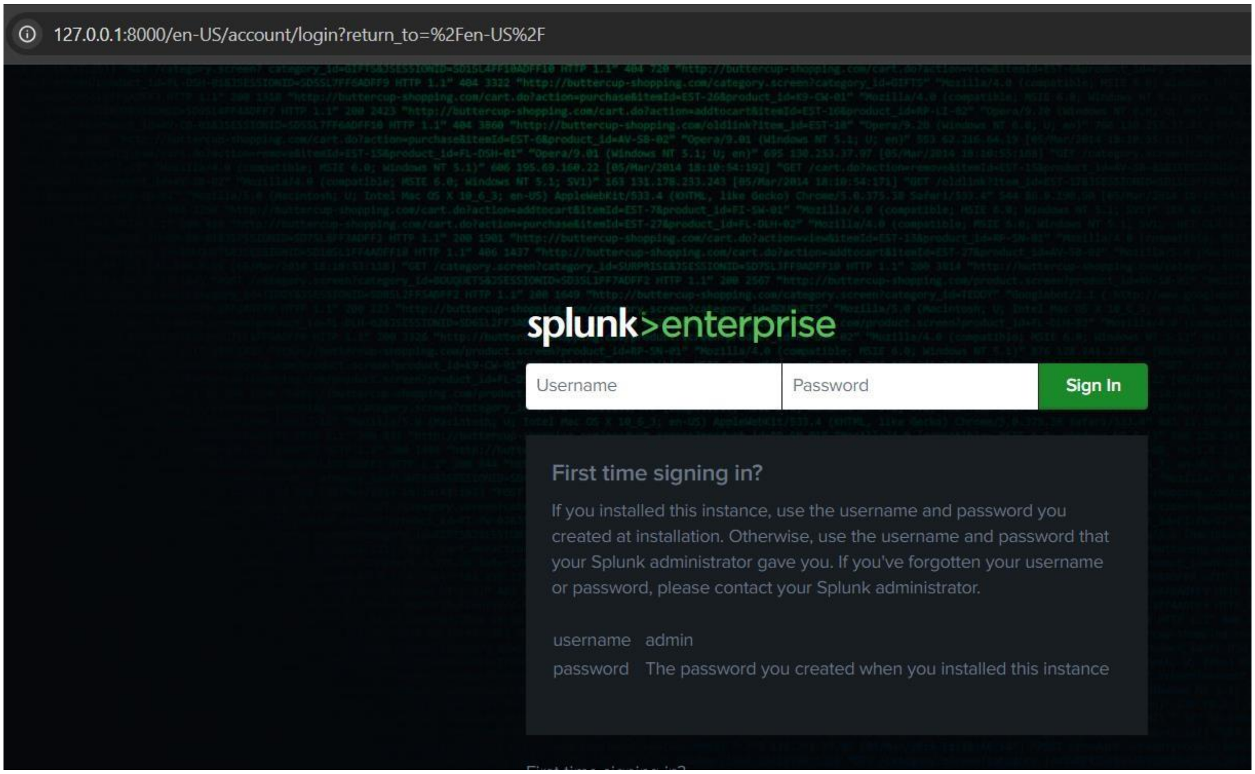
- Choose windows and download the msi
- Once downloaded, locate the .msi file and run the installer
- During installation, you'll be asked to set the installation directory, accept the license agreement, and choose to create shortcuts.

Set Splunk Admin Credentials:

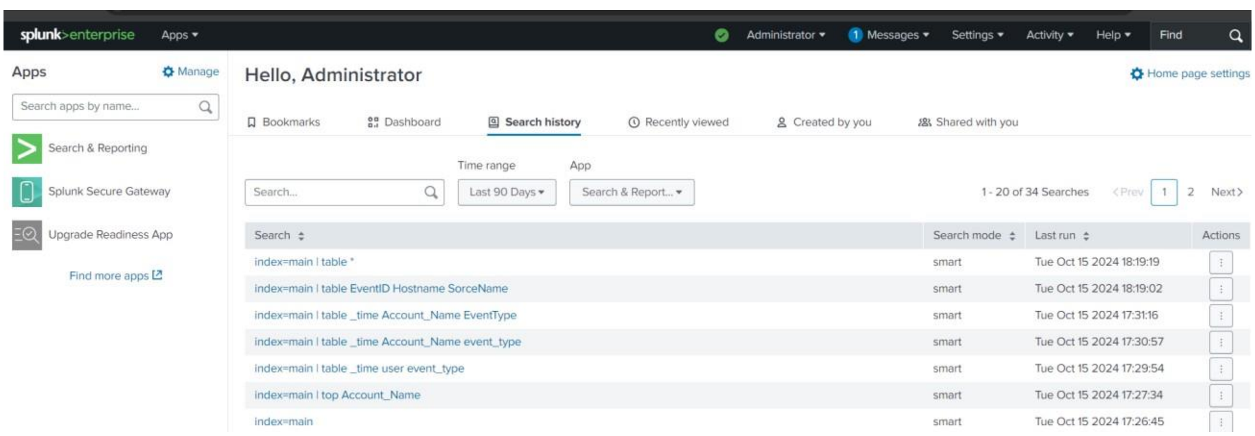
- You’ll be prompted to set a username and password for the Splunk admin account during the installation. This will be used to log into Splunk’s web interface. You can open it



Enter the username and password you have written



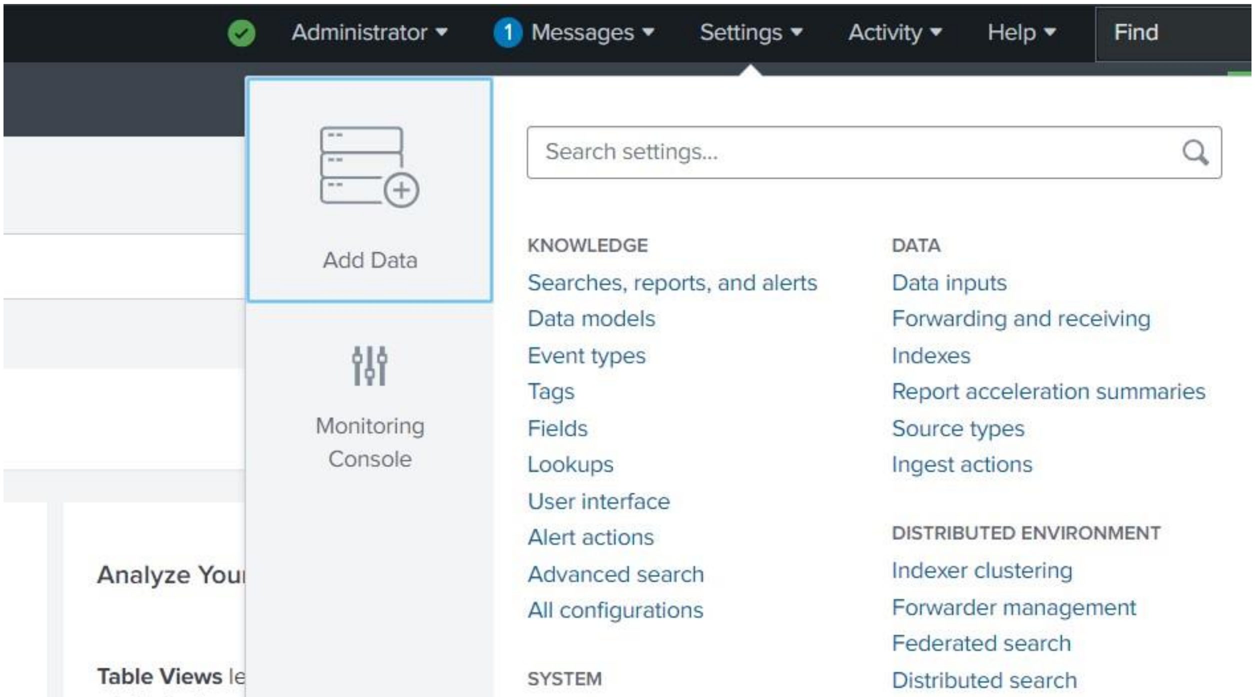
Then it will open



To add network logs

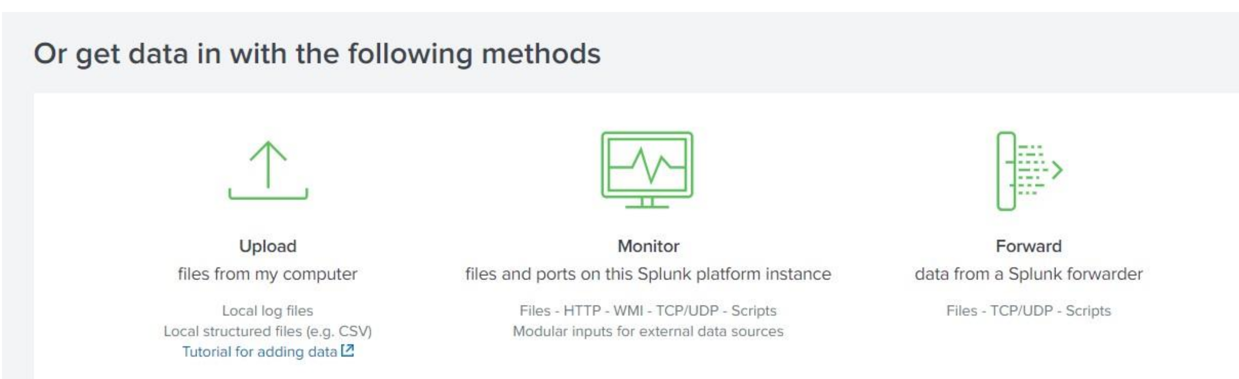
1.Add data

- In the Splunk home screen, click on "Settings" in the top-right corner.
- Under "Data", click "Add Data".



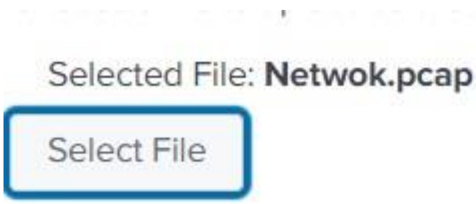
2.select data source

- You will see three options: Upload, Monitor, and Forward.
- Choose "Upload" if you want to upload a network log file manually



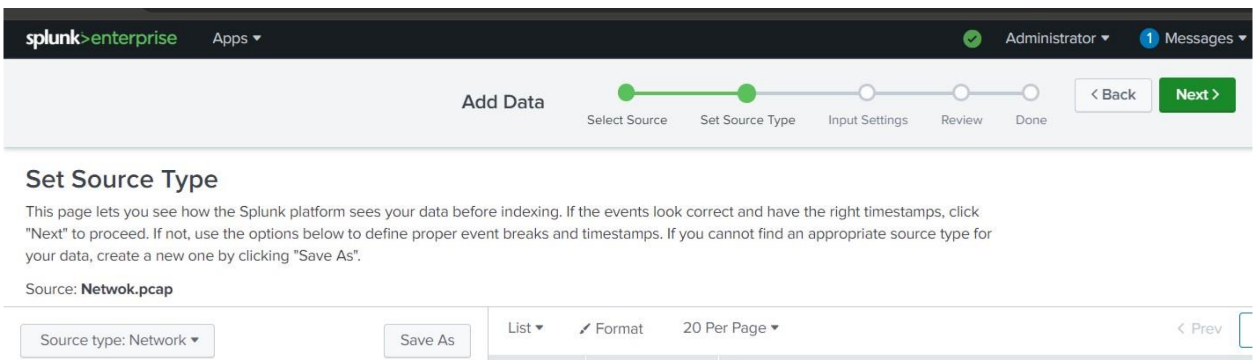
3.Upload the Network Log File:

- Click "Select File" and browse your computer for the network log file you want to upload.
- Once the file is selected, click "Next".



4. Source Type:

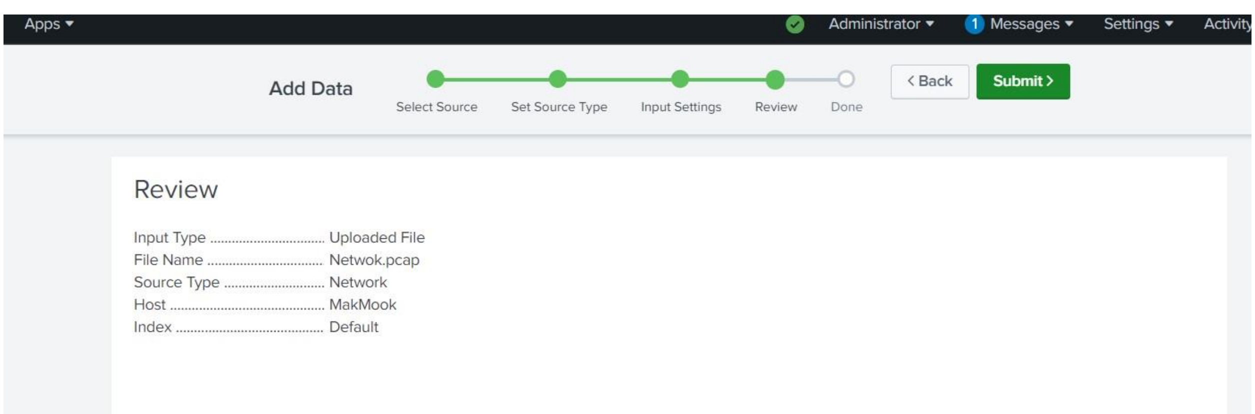
- Splunk will try to automatically detect the source type
- If Splunk does not correctly identify the source type, manually choose one that best fits your data



- Click "Next" once the source type is selected.

5. Set Host, Index, and Sourcetype:

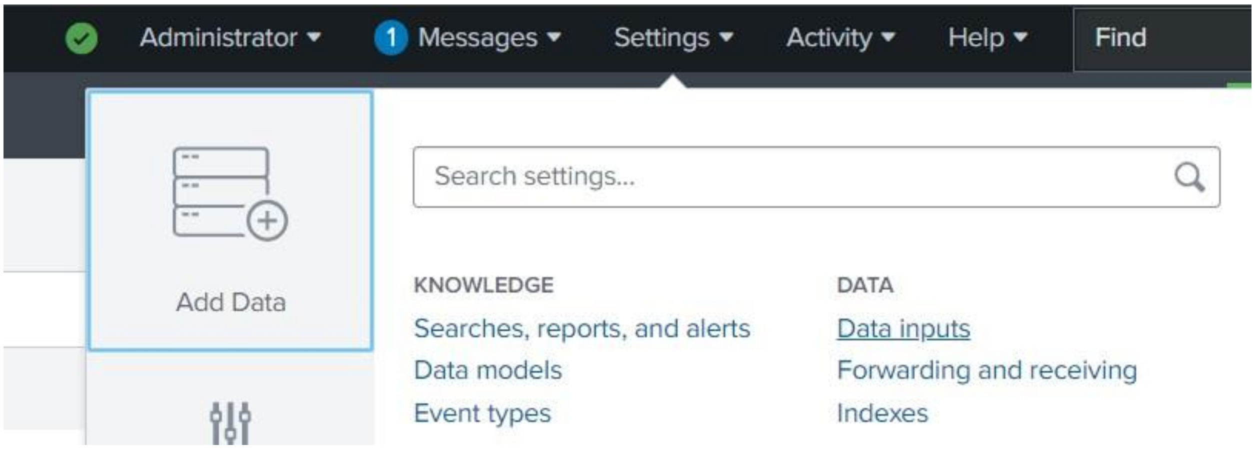
- **Host:** Identify the host from which the logs were generated. You can manually specify or let Splunk assign a host.
- **Index:** Select an index or create a new one to store your data. By default, you can use the "main" index.
- **Sourcetype:** Ensure the correct sourcetype is selected.
- After reviewing these settings, click "Review" and then "Submit".

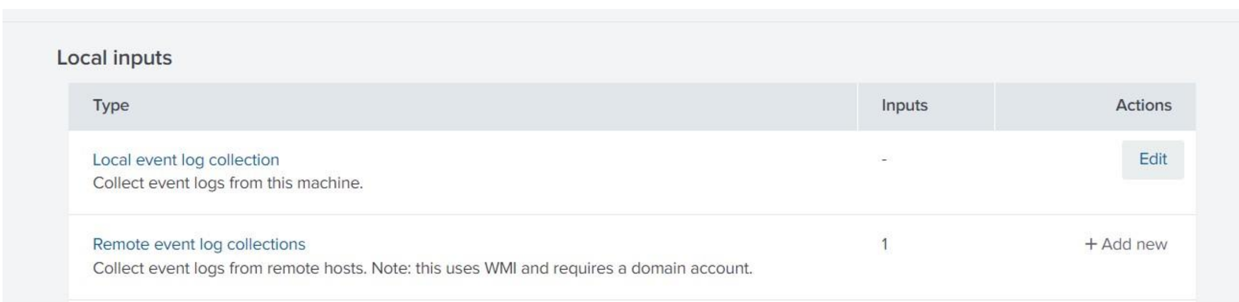


Data Uploaded:

- Your network log data will now be uploaded into Splunk and available for searching and analysis.

To add your logs





Choose local event log collection and choose which logs you want to add then save

Name ? localhost

Logs

Available log(s) Add all »		Selected log(s) « Remove all
Application ✓	>	Application
Dell ✓	<	Dell
DirectShowFilterGraph		ForwardedEvents
DirectShowPluginControl		General Logging
Els_Hyphenation/Analytic		HardwareEvents

Select the Windows Event Logs you want to index from the list.

i	Time	Event
>	10/15/24 2:51:31.000 PM	(see below)

```
10/15/2024 02:51:31 PM
LogName=Security
EventCode=5379
EventType=0
ComputerName=MakMook
SourceName=Microsoft Windows security auditing.
Type=Information
RecordNumber=5178886
Keywords=Audit Success
TaskCategory=User Account Management
OpCode=Info
Message=Credential Manager credentials were read.

Subject:
        Security ID:            S-1-5-21-2554599862-2962239839-24097432-1001
        Account Name:           emanM
        Account Domain:         MAKMOOK
        Logon ID:               0x13BF13A2
        Read Operation:         Enumerate Credentials

This event occurs when a user performs a read operation on stored credentials in Credential Manager.
```

Collapse

host = MAKMOOK | source = WinEventLog:Security | sourcetype = WinEventLog:Security